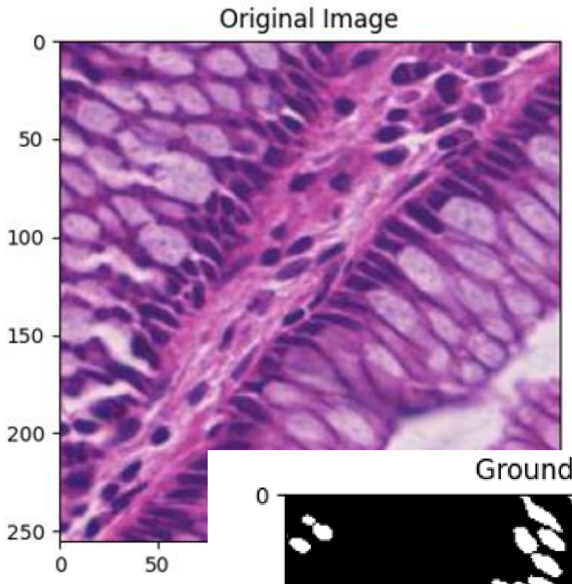
COLON NUCLEI SEGMENTATION USING IMAGE ANALYSIS AND DEEP LEARNING

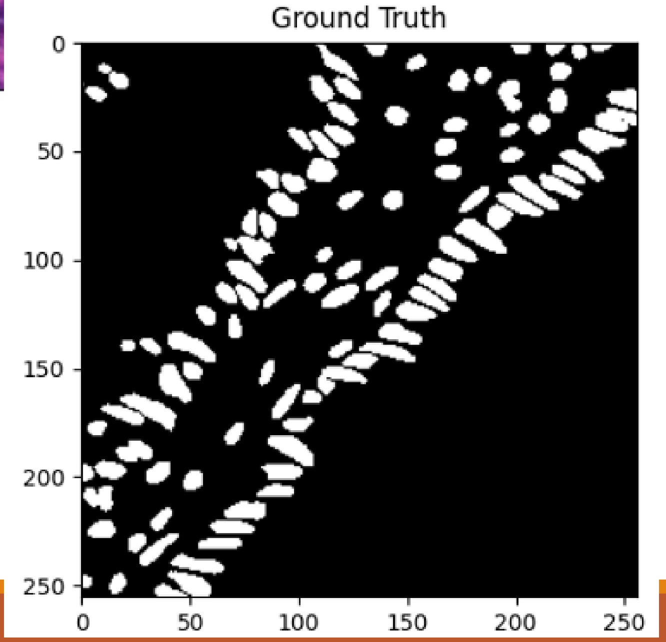
NAME: YEDUPATI HARSHA VARDHAN
ROLL NO: 22EC01007
SUPERVISOR: DR. HIMANSHU PRAMOD PADOLE

Background
- Histopathological image analysis plays a crucial role in colon cancer diagnosis
- Nuclei segmentation helps in quantitative analysis, grading, and treatment planning
- Manual nuclei segmentation is slow and difficult in densely packed tissue regions.

Challenges
- Overlapping nuclei
- Irregular boundaries
- Dense cellular clusters
- Staining variations

Objective
Develop and compare traditional image processing and deep learning models for:
- Accurate nuclei segmentation
- Nuclei counting
- Performance evaluation





DATASET AND PREPROCESSING

CoNIC Dataset

- **Dataset Used: CoNIC**
- Colon Nuclei Identification and Counting Dataset
- Contains **4981 histopathology images**
- Image size: **256 × 256 × 3**
- Includes:
  - Original histology images
  - Instance segmentation masks
  - Nuclei annotations

Data Preparation

Raw Histopathology Images

↓

Normalization

$$I_{norm} = \frac{I}{255}$$

Pixel values scaled to range [0,1]

Improves training stability

↓

Binary Mask Generation

$$Mask(x,y) = \begin{cases} 1, & label(x,y) > 0 \\ 0, & otherwise \end{cases}$$

Converts instance masks into binary segmentation masks

Foreground → nuclei

Background → non-nuclei regions

↓

Input Pair Formation

(Image, Mask) pairs created for model training

Train-Test Split

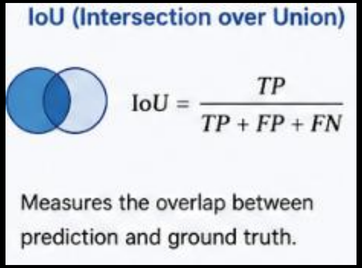
Dataset Split

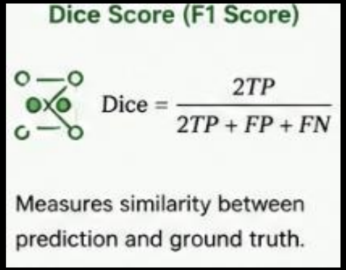
- Total Images: **4981**
- Training Images: **3984**
- Testing Images: **997**
- Split Ratio: **80:20**
- Random State: **42**

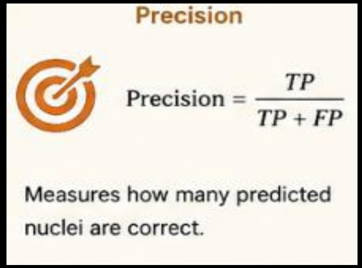
Why Split?

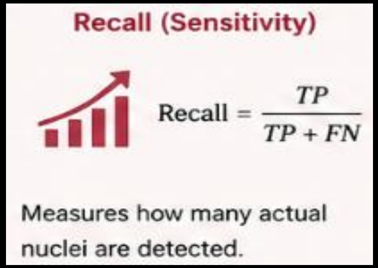
- Training set → model learning
- Testing set → final performance evaluation
- Prevents overfitting bias

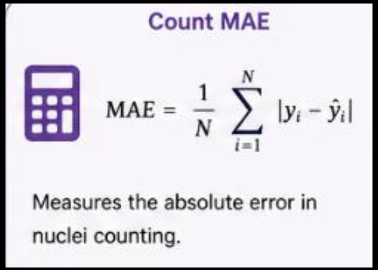
Evaluation Metrics











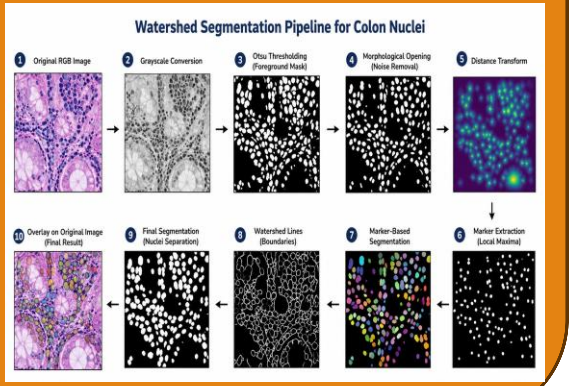
METHODOLOGY: BASELINE SEGMENTATION MODELS

WATERSHED SEGMENTATION

Traditional Image Processing Baseline

Pipeline:

→ Grayscale conversion
→ Gaussian smoothing
→ Distance transform
→ Marker generation
→ Watershed segmentation
→ Nuclei extraction



U-NET

U-Net Architecture

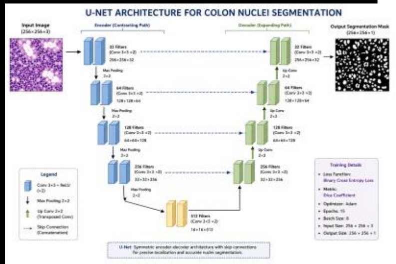
- Encoder-decoder architecture
- Skip connections preserve spatial features
- Binary segmentation output

Loss:

$$BCE = -[ylog(p) + (1-y)log(1-p)]$$

Metric:

Dice Score



U-NET++

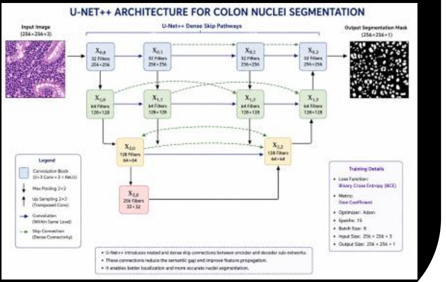
U-Net++ Architecture

- Nested skip pathways
- Reduces semantic gap between encoder-decoder features
- Better feature fusion than U-Net

Loss:

BCE Binary Cross Entropy

Metric: Dice Score



SA-NET

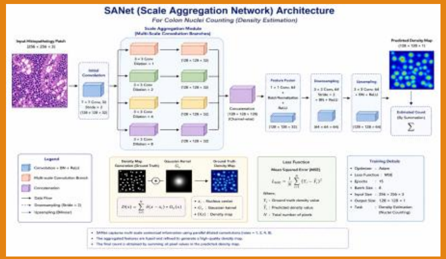
Scale Aggregation Network

- Multi-scale convolution branches
- Kernel sizes: 3×3, 5×5, 7×7
- Generates density maps for nuclei counting

Loss:

MSE Loss

$$MSE = \frac{1}{N}\sum(y-\hat{y})^2$$



METHODOLOGY: MODEL ENHANCEMENTS

ATTENTION U-NET++

Attention U-Net++ Architecture

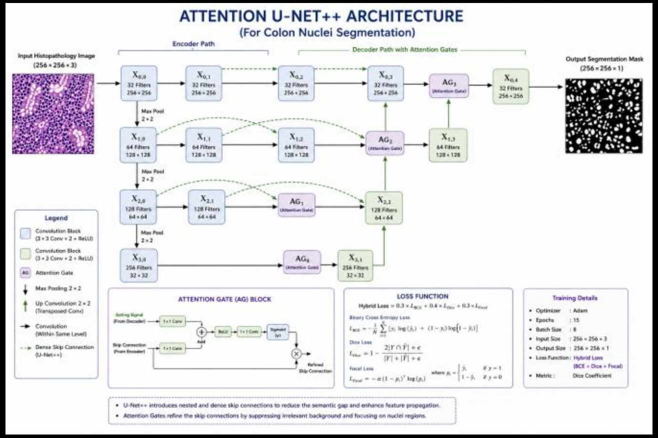
- Attention gates added to skip connections
- Suppresses irrelevant background features
- Improves nuclei localization
- Enhances segmentation boundaries

Hybrid Loss Function

$$L = 0.3(BCE) + 0.4(DiceLoss) + 0.3(FocalLoss)$$

Why Focal Loss?

Handles class imbalance between nuclei and background pixels.



MULTISCALE ATTENTION + JPFM MODEL

Key Enhancements

- Multi-scale feature extraction
  - 1×1 convolution
  - 3×3 convolution
  - 5×5 convolution
- Attention refinement
- Better feature selection

JPFM bottleneck

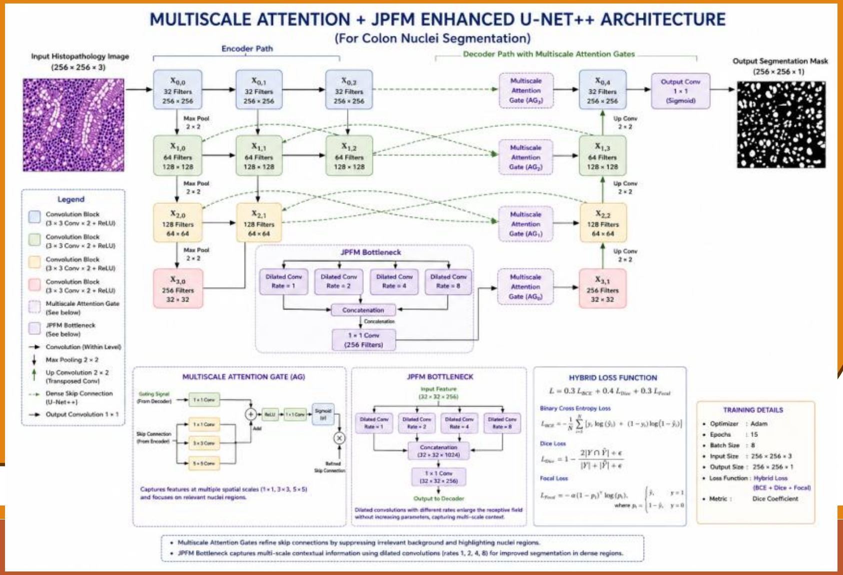
- Dilated convolutions
- Better contextual learning

Why Final Model Performed Better?

- Better feature extraction
- Better localization
- Improved handling of dense nuclei clusters

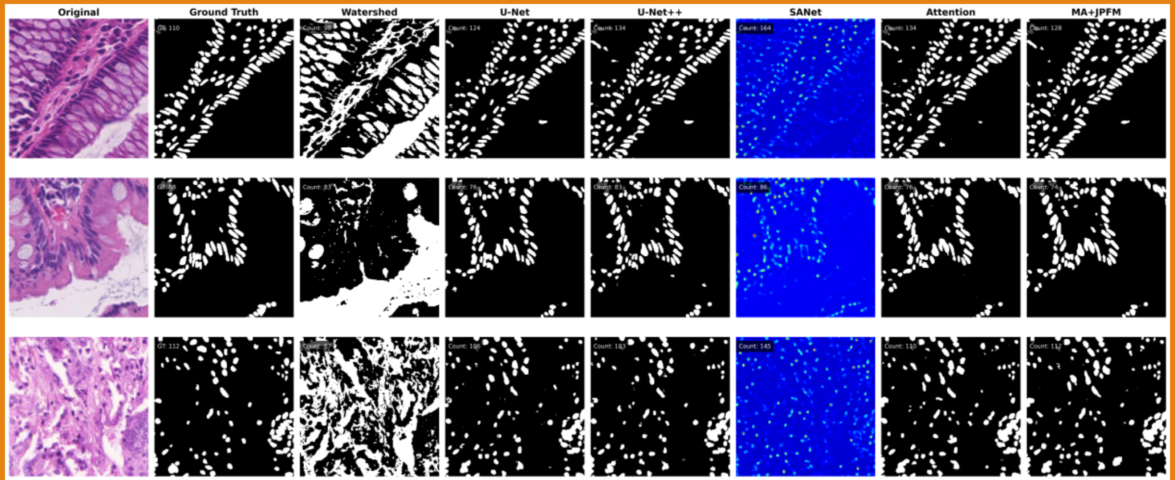
Limitations Observed

- Missed small nuclei
- Poor multiscale feature extraction
- Boundary inaccuracies
- Difficulty in clustered nuclei regions



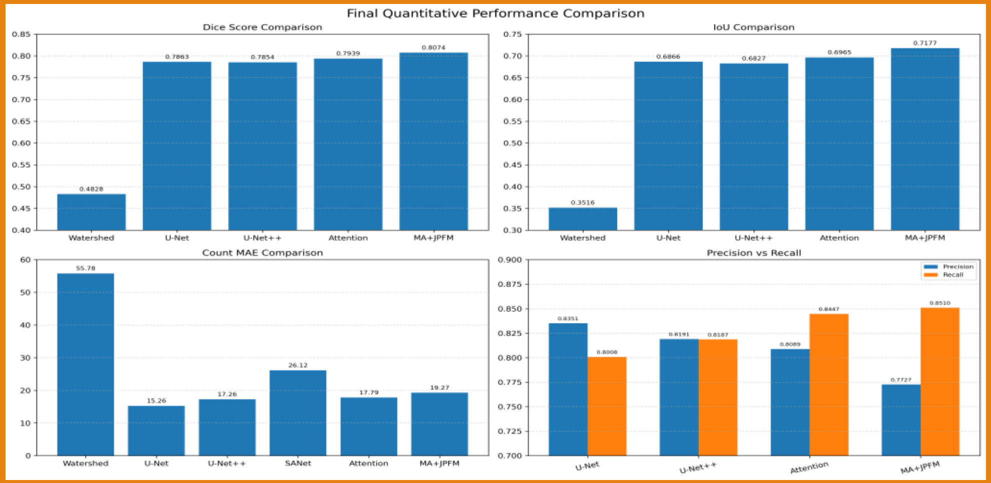
EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

Qualitative Results



Visual comparison of segmentation outputs and nuclei count estimation.

Quantitative results

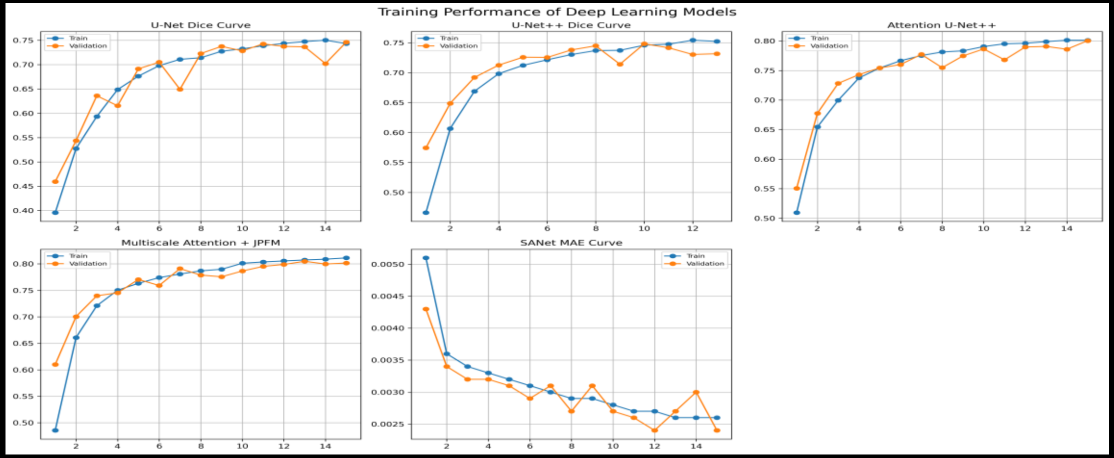


Final model achieved highest IoU and Dice score.

Best Performing Model:
MultiScale Attention + JPFM

- Highest IoU
- Highest Dice
- Best Recall

Training Analysis



Final Performance Table

Model	IoU	Dice	Precision	Recall	Count MAE
Watershed	0.3516	0.4828	–	–	55.78
U-Net	0.6866	0.7863	0.8351	0.8008	15.26
U-Net++	0.6827	0.7854	0.8191	0.8187	17.26
SANet	–	–	–	–	26.12
Attention U-Net++	0.6965	0.7939	0.8089	0.8447	17.79
MultiScale + JPFM	**0.7177**	**0.8074**	0.7727	**0.8510**	19.27

CONCLUSION AND FUTURE SCOPE

Conclusion

Key Findings

- Watershed segmentation struggled with overlapping nuclei.
- Deep learning models significantly improved segmentation performance.
- U-Net++ improved feature fusion.
- SANet enabled nuclei counting through density estimation.
- Attention mechanisms improved localization
- Final model achieved best overall segmentation performance.

Applications

- Computer-aided pathology analysis
- Cancer diagnosis and grading
- Automated nuclei counting
- Tissue morphology analysis
- Clinical workflow automation

Future Scope

Future Work

- Vision Transformer-based segmentation
- Whole slide image analysis
- Multi-class nuclei classification
- Real-time clinical deployment
- Foundation model integration

References

- Ronneberger et al., **U-Net: Convolutional Networks for Biomedical Image Segmentation**, MICCAI, 2015
- Zhou et al., **U-Net++: A Nested U-Net Architecture for Medical Image Segmentation**, 2018
- Oktay et al., **Attention U-Net**, 2018
- CoNIC Challenge Dataset, 2022
- Qi Liu et al., **Multi-module UNet++ for Colon Histopathological Segmentation**, Scientific Reports, 2025